

بسم الله الرحمن الرحيم

Intestinal Protozoa 2- Giardia Lamblia

First Course Lecture 4

Giardia lamblia

objective

- **Disease:** Giardiasis.
- **Causative agent:** *Giardia lamblia*.
- **Infective stage:** mature cyst from patient's stool or carrier.
- **Route of entry:** Orally
- **Source of transmission:** by fecal- oral route (from patient's stool)
- **Habitat:** small intestine mainly in duodenum and jejunum.
- **Host:** only intestine of humans
- **Diagnostic stage:** cyst and trophozoite stage in patient's stool or carrier.
- Giardiasis more prevalent in children.
- **Geographical distribution:** and the percentage of infection increases among poor communities.

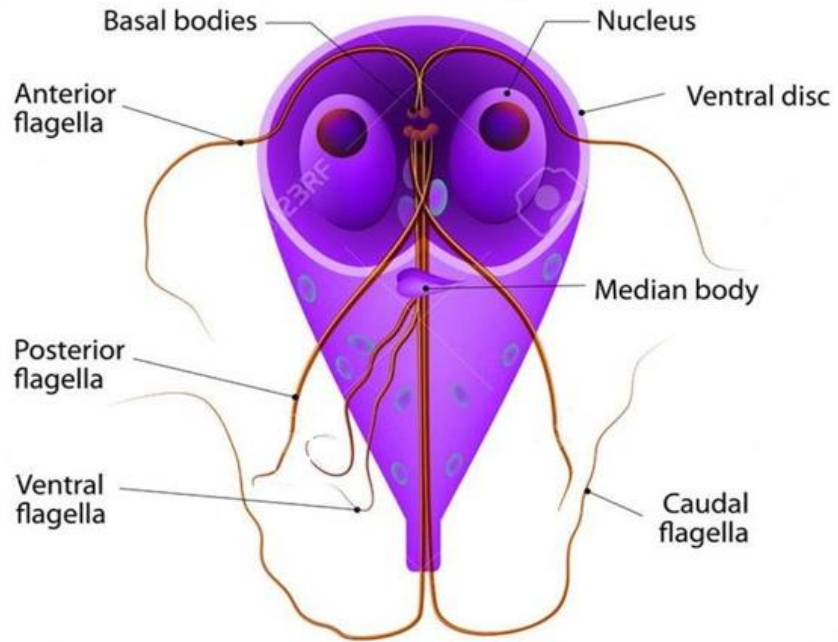
Introduction

Giardia lamblia was first observed by Antonie Von Leeuwenhoek in 1681 in his own stools, and in 1915 Prof. Guard (Paris) and Prof. Lambie (Prague) were to give a name to this protozoal parasite and also gave a detailed description of it. *Giardia* is a genus of anaerobic flagellated parasite of the phylum melamonada that colonie and reproduce in the mucosa of human small intestine and causes intestinal giardiasis.

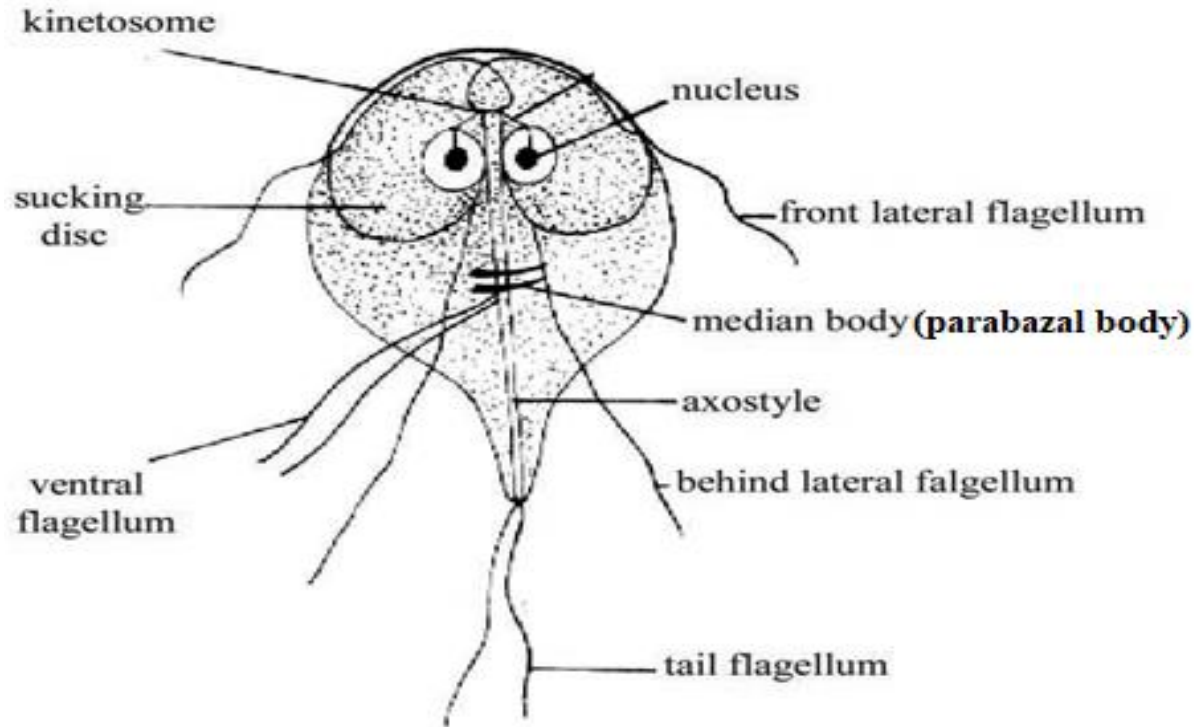
It is a protozoal parasite found in mucosal small human intestine mainly in duodenum and upper jejunum and does not invade the tissue or penetrate the wall of small intestine, it is the most common protozoan pathogen worldwide in distribution. Epidemically is very high in area with low sanitation. ***Giardia lamblia*** protozoal adhered to intestinal epithelium (mucus) by means of sucking disc.

Morphology

Giardia lamblia have two stages, trophozoite and cyst stage, trophozoite is a heart racket or pyriform shaped, measures 15x9 μm wide and thick, rounded anteriorly and pointed posteriorly, dorsally convex and ventrally with concave two sucking discs in front of body with each one has one spherical nucleus with nuclei which helps in its attachment to the human intestinal mucosa. *Giardia lamblia* it have two round nucleus with two nuclei, four pairs of flagella with basal bodies of long free flagella (anterior, lateral caudal, posterior, and ventral) and two parabasal (median) sausage shaped body ,one pair of axostyles running along the midline and motile by free flagella. *Giardia* cyst are relatively resists to disinfection like chlorination and UV light



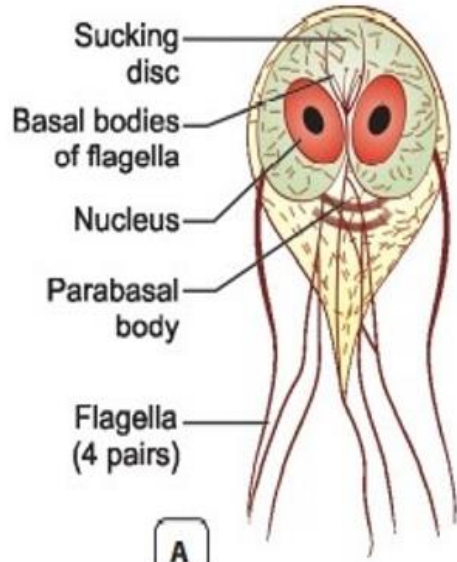
Morphology of trophozoite of *Giardia Lamblia*



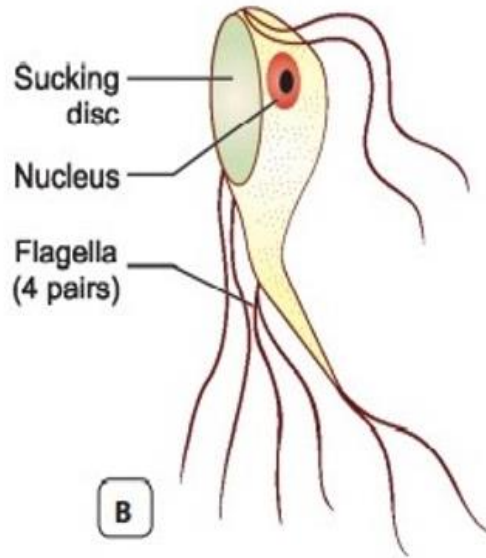
Schematic diagram to trophozoite of *Giardia Lamblia*



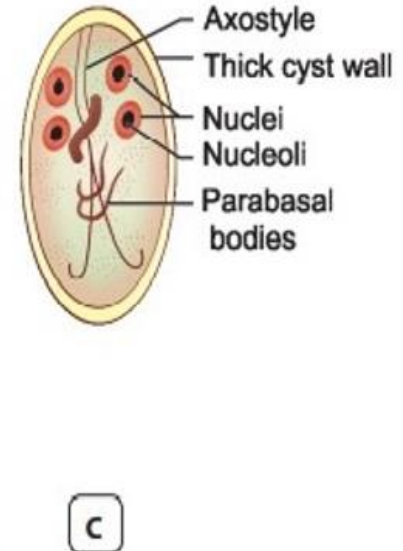
Mature cyst stage is an infective stage, it is small, ovoid shape, with thick wall, and its size measuring 12x8 μm , surrounded (covered) by thin smooth hyaline cyst wall, the cyst internal contains a cytoplasm with small granules and parabasal body (no free flagella) and endo-axostyle, the mature cyst is characterized by the presence of four small spherical nuclei with four central nucleoli site in one end and includes axostyle and parabasal body



Trophozoite



Ventral view



Quadrinucleated cyst

Morphology of *Giardia Lamblia* and cyst stage

A&B. Trophozoite of *giardia lamblia*, C. Mature cyst stage

Life cycle

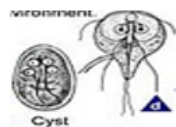
Giardia lamblia need one host (human) to complete the life cycle, The infections start when human ingest mature cyst (infective stage) in contaminated food, and water from patient's stool or carriers, or by the fecal-oral route, cysts can survive several months in cold water, cyst stage in stomach is resistant to gastric juices and pass to cavity of small intestine, after half an hour of ingestion, the excystation takes place in duodenum, the cyst hatches or the cyst wall get damaged and each cyst liberate two trophozoites, which grow and multiply successively by longitudinal binary fission and increase in number and colonized in mucosa of upper part of the small intestine mainly duodenum and jejunum, feeding by pinocytosis. The trophozoites may swim freely or attach to the sub-mucosal epithelium by their ventral suction discs.



Trophozoite of ***Giardia lamblia*** does not invade the tissue of human small intestine.

The trophozoite during unfavorable condition, encystment may occur in lower portion in cavity of small intestine and usually in colon and the cysts escape with stool.

Some of trophozoites may pass with stool and remain viable in soil or waters for several weeks, up to 200.000 cysts may escape per one gram of patients stool, sometime the trophozoites colonized and cover the bile duct and pancreas.

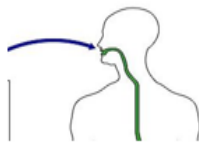


**Mature Cyst stage
(infective stage)**

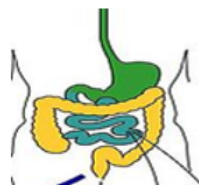
**Cyst escape with
patients stool**

Schematic diagram
of life cycle of
giardia lamblia in
human host

**Ingesting by
human**



**In stomach
resistant**



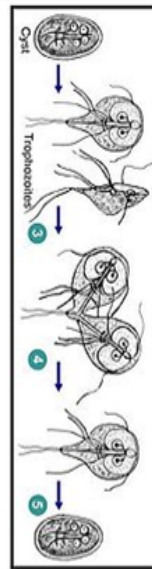
**colonized and cover
the bile duct and
pancreas**

**Escape with
patients stool**

**Trophozoite encystment in lower part of cavity of
small intestine (usually in colon)**



**Excystation of Cyst in
cavity of small
intestine**



**Liberate two small
Trophozoite and
grow and multiply**



**Trophozoite attach and colonized and
growth in mucosa of duodenum and
jejunum and submucosal epithelium**

Giardia Lamblia life cycle

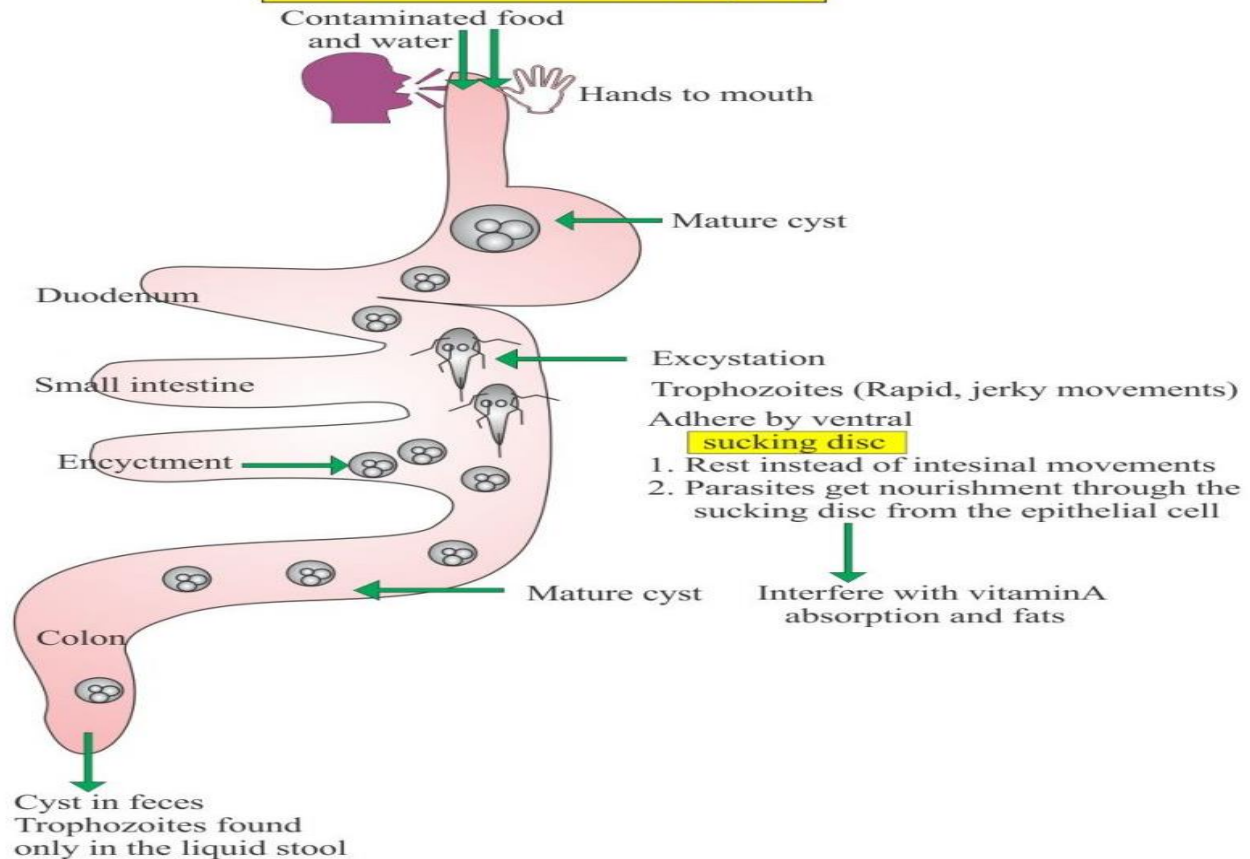


Diagram of life cycle of *giardia lamblia*

Pathogenesis and clinical features

Giardia lamblia causes giardiasis is typically seen within the crypts of duodenal and jejunal mucosa, it does not invade the tissue of intestine, but remains lightly adhered to intestinal epithelium by means of sucking disc, It is asymptomatic in more than 50% of giardiasis cases, but they are carriers, some cases of giardiasis may lead to diarrhea with mucus without blood, due to mucosal ulceration in small intestine. In carriers they may have diarrhea followed by constipation and gases. When ***Giardia lamblia*** infection causes enterocytes damage and loss of brush border of the epithelial cells of the intestine that leads to shorting of microvilli, and altered epithelial. Barrier function colonized in the gallbladder (bile) and pancreas duct lead to malabsorption of fat and vitamin, it causes acute diarrhea and low grade fever, nausea, and anorexia. Incubation period about two weeks (1-3 weeks) from infection.



Complication and clinical symptoms

Occasionally patients with giardiasis will have long term complications such as reactive arthritis, irritable bowel syndrome (IBS), recurring diarrhea, and fatty diarrhea (steatorrhea) that can last more than one year, and may delay physical and mental growth, slow development, and causes malnutrition.

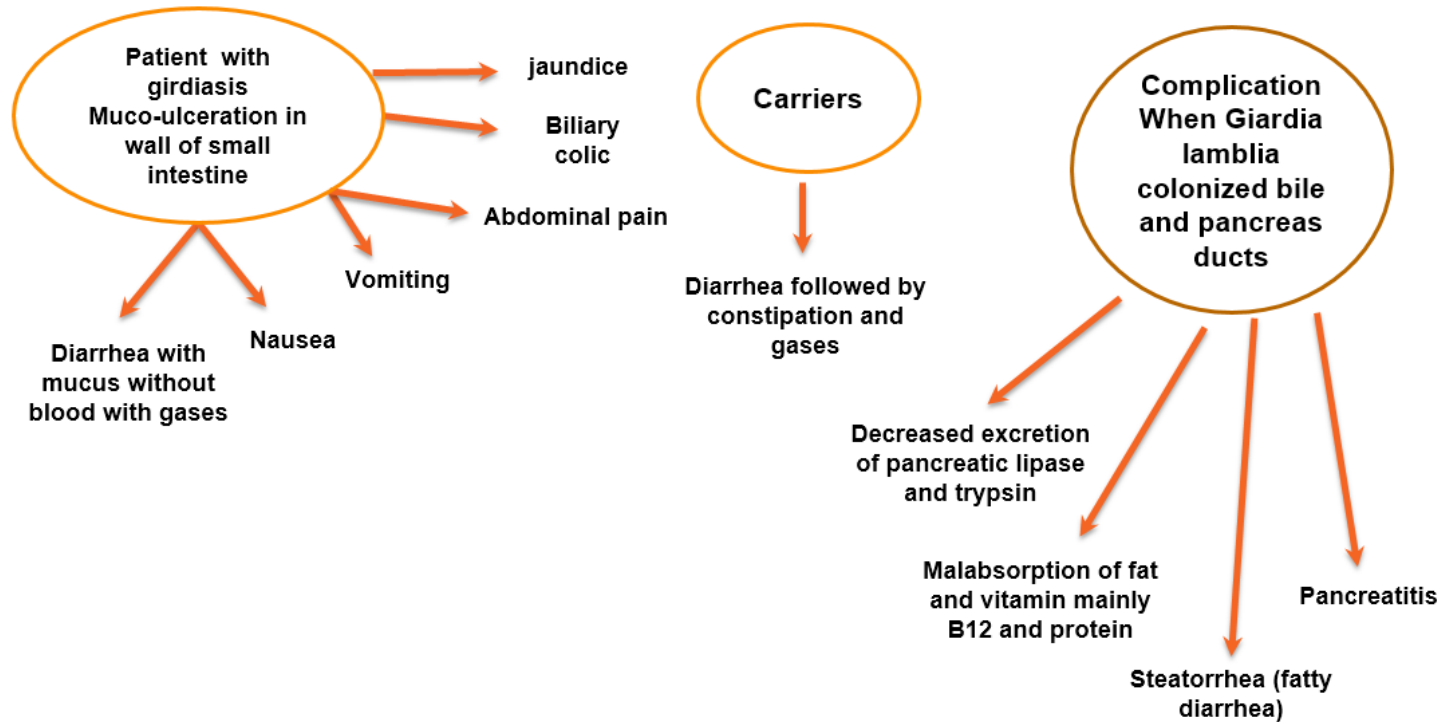


Complications may include:

1. Decrease excretion of pancreatic lipase mainly in pediatric patients.
2. Decrease of Lag secretion from mucoepithelial cells, has been considered a factor of being predisposed to Giardiasis.

The clinical symptoms of giardiasis include nausea, vomiting, diarrhea, gas, vary common of pancreatitis, foul-smelling, greasy stool that can float, stomach cramps pain and dehydration, right abdominal pain with weakness, loss of weight, loss of appetite, anorexia, and causes biliary colic and jaundice.





Schematic diagram of disease and clinical symptom of giardiasis



Laboratory diagnosis

Stool examination:

I. Macroscopic examination: may have an offensive odor, pale colored, fatty, diarrhea (steatorrhea) and float in water. May be profuse watery and later become greasy (fatty) and foul smelling (odor) with no blood.

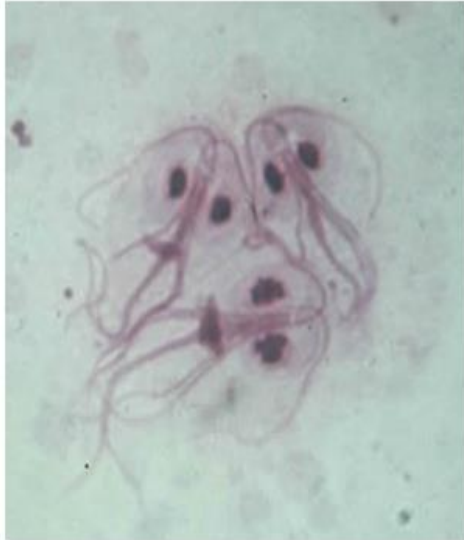
II. Microscopic examination: is usually based on the recovery of typical cyst stage and trophozoites in patient's stool or carrier by normal saline and solution of iodine wet preparation.

III. Entero – test (string test): this test is a commonly available diagnostic method for the recovery of upper gastrointestinal tract, can be used for giardiasis, may be the stool need concentration techniques like formal-ether.



Serological test: antigen and antibody detection by (ITF) test or by Elisa test to detect anti body or antigen against Giardia.

Molecular diagnosis: By DNA probes and by polymerase chain reaction (PCR) have been used to demonstrate protozoal genome in patient stool.



A



B



C

Microscopic examination of *Giardia lamblia* in patient stool

A&B. Trophozoite stage, C. Cyst stage



Treatment

The drug of choice or the best treatment are **metronidazole (Flagyl)** 250 mg orally 3 times for 5-7 days or 2 g daily for 3 days with **vitamin B12**, with **tinidazole** 2 g single dose. The pregnant females can be treated by **paromomycin** and **oral aminoglycoside**.



Prophylaxis

At general giardiasis can be prevented by proper disposal of waste water and patient's stools, disinfection to water by chlorination or boiling of water, hand washing before eating, and treating the human carriers. The best prevention is by following measures of proper disposal of waste water and practice of personal hygiene like hand washing before eating and proper disposal of diapers, chlorination of water is effective for inactivate cysts, Boiling of water or filtration by special membrane filters are required.



Thank you for listening, any questions?